Day 14 Report

Today's session began with a detailed reading and overview of **Microsoft Foundry** documentation. We explored the project structure, key components, and how different sections are organized within the platform.

Morning Session

1. Images and Image Management

We learned about:

- The definition of an image in cloud environments
- Best practices before creating an image (such as cleaning temporary files and emptying the recycle bin)
- Why system cleanup is important for optimized image creation

2. Representations and Cloud Labs Overview

We had an overview session on the **Cloud Labs** tool, where we explored:

- Multi-agent systems and their use cases
- Why multi-agent architecture is implemented
- A newly introduced feature called **Embedded Shadow**, including:
  - Its purpose
  - The problems it solves
- Logon scripts:
  - What they are
  - Why they are used in system configuration and automation

Afternoon Session – Networking Deep Dive

In the afternoon, we had an in-depth session on networking fundamentals, particularly in **Microsoft Azure**.

Topics Covered:

- Basics of networking
- IP addresses
- Subnet masks
- DNS and how it works
- Virtual Networks (VNets) and their functionality
- Securing Top-Level Domains (TLDs)
- DHCP (Dynamic Host Configuration Protocol)
- Types of DNS records

Types of DNS Records Discussed:

- A Record
- AAAA Record
- CNAME Record
- MX Record
- TXT Record
- NS Record
- PTR Record

Practical Tasks Completed

Task 1: Virtual Machines and Load Balancer

- Launched two Virtual Machines (VMs) in a single Virtual Network
- Attached both VMs to a Load Balancer
- Configured a user script (HTML-based) on VM1
- Verified functionality by pinging the Load Balancer IP
- Successfully received the webpage hosted on VM1

Task 2: Traffic Manager

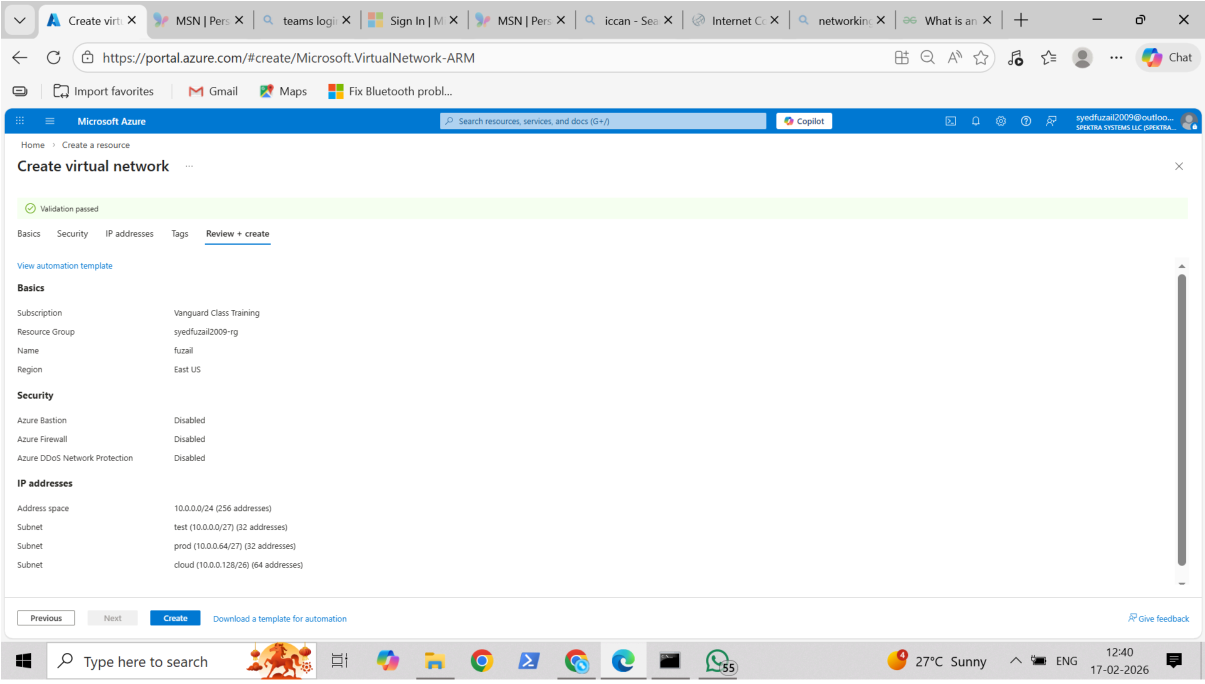
We also worked on understanding **Azure Traffic Manager**, including:

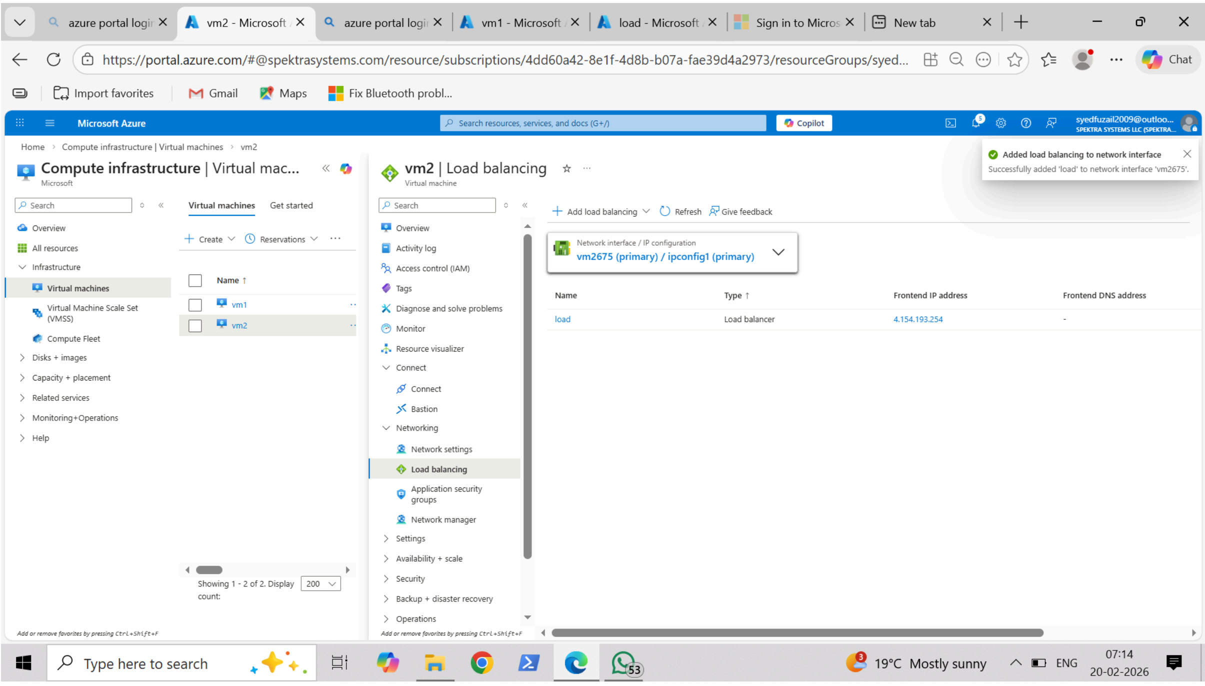
- What it is
- How it works
- Its purpose in traffic distribution
- Differences between Load Balancer and Traffic Manager

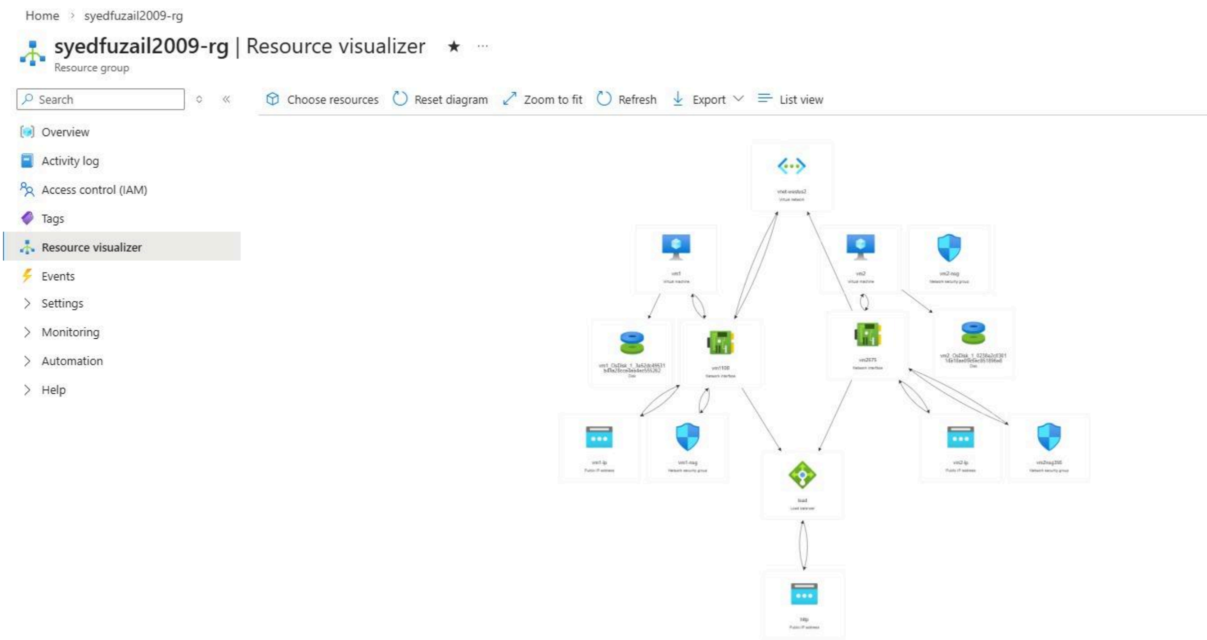
Conclusion

Today's session strengthened our understanding of:

- Microsoft Foundry documentation
- Cloud Labs and multi-agent architecture
- Image management best practices
- Networking fundamentals in Azure
- DNS, DHCP, Load Balancing, and Traffic Manager


https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM
Microsoft Azure
Home > Create a resource
Create virtual network
Validation passed
Basics
Security
IP addresses
Tags
Review + create
View automation template
Basics
Subscription	Vanguard Class Training
Resource Group	syedfuzail2009-rg
Name	fuzail
Region	East US
Security
Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled
IP addresses
Address space	10.0.0.0/24 (256 addresses)
Subnet	test (10.0.0.0/27) (32 addresses)
Subnet	prod (10.0.0.64/27) (32 addresses)
Subnet	cloud (10.0.0.128/26) (64 addresses)
Previous
Next
Create
Download a template for automation
Give feedback
Type here to search
27°C Sunny
ENG
12:40
17-02-2026


https://portal.azure.com/#@spektrasystems.com/resource/subscriptions/4dd60a42-8e1f-4d8b-b07a-fae39d4a2973/resourceGroups/syed...
Microsoft Azure
Home > Compute infrastructure | Virtual machines > vm2
Compute infrastructure | Virtual mac...
Microsoft
Overview
All resources
Infrastructure
Virtual machines
Virtual Machine Scale Set (VMSS)
Compute Fleet
Disks + images
Capacity + placement
Related services
Monitoring+Operations
Help
Virtual machines
Get started
Create
Reservations
Name
vm1
vm2
Showing 1 - 2 of 2. Display count:
200
vm2 | Load balancing
Virtual machine
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Connect
Bastion
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings
Availability + scale
Security
Backup + disaster recovery
Operations
Add load balancing
Refresh
Give feedback
Network interface / IP configuration
vm2675 (primary) / ipconfig1 (primary)
Name	Type	Frontend IP address	Frontend DNS address
load	Load balancer	4.154.193.254	-
Added load balancing to network interface
Successfully added 'load' to network interface 'vm2675'.
Type here to search
19°C Mostly sunny
ENG
07:14
20-02-2026


Home > syedfuzail2009-rg
syedfuzail2009-rg | Resource visualizer
Resource group
Search
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Monitoring
Automation
Help
Choose resources
Reset diagram
Zoom to fit
Refresh
Export
List view
vnet-westus2
vm1
vm2
vm2-nsg
vm1-ip
vm1-nsg
vm2-ip
load